OsmAnd shade plugin project

Plugin Documentation

This is a small documentation about custom plugins in OsmAnd, which uses our examples made throughout our project.

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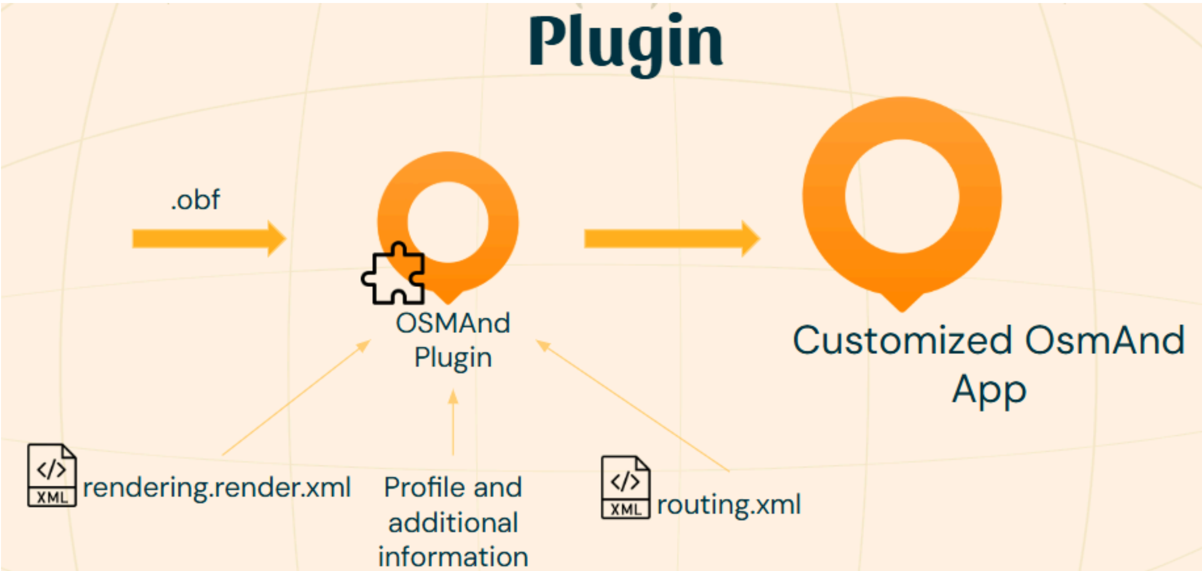
1. General informations about plugins

In OsmAnd, we can import a .osf file containing information, which allows us to customize the app. This is what we called a custom plugin. To have a global view of a plugin, here is a simple picture describing it :



Let's delve into the details.

a) The required and specific files

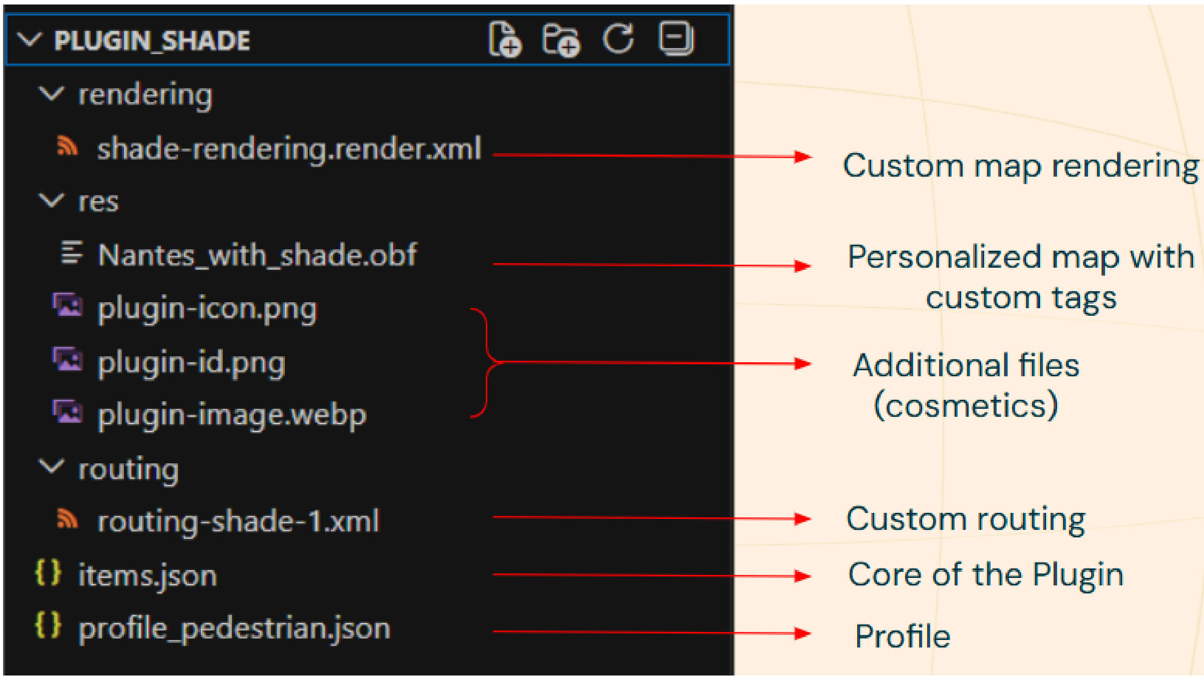
To create a simple plugin that add a custom profile in the app, we need 2 files : a file for setting up the plugin, called items.json and a file for specific information about the new profile, called <as you want>.json. In our example, we will use items.son and profile_pedestrian.json.

Note that a plugin is a static customization of the app (not dynamic), that means we can't modify dynamically what the app does with a plugin, for example we can't use the current time for a calculation by using a plugin. That also means all files in the plugin will be static files (xml, json, png, …).

Then we can add optional documents to customize our new profile in the resource folder (res). For example, we added pictures to have an icon and a background for our plugin.

Finally, we can add two files to modify the app concretely : the rendering.render.xml and the routing.xml files. The rendering file allows you to modify how the map looks like in the app, while the routing file allows you to modify a route/trajectory by modifying the route calculation when using the app. Moreover, these files can use tags from the OsmAnd app, and it is possible to add personal tags to a custom OsmAnd app. Usually, a customized map (.obf) is created with custom

tags and put in the plugin. This allows the rendering and routing files to use the custom tags.
Here is a simple picture describing the plugin with all these elements :



b) How to make the .osf

The plugin should be put into the app as an .osf file. To simply make a plugin :
1- create a folder with all the previous elements (or only those you need) on your computer.
2- Select all these elements from the folder (do not select the folder itself) and zip them. For example, with the picture above : we would select the 3 folders rendering, res and routing, with the 2 files items.son and profile_pedestrian.json, and zip them.
3- Rename the .zip as .osf and now you have your plugin file.

c) Specific to each files

i) Items.json

This file is the only required file, it is the core of the plugin. You will find information about it in the readme from the “plugin” folder on the GitHub of our project.
Note that its configuration is really important, it is the first file read by the app and it allows you to download all the other files added in the .osf (the

additional ressources, your custom map and your custom rendering and routing).

ii) Others

You will find some information about the other files that can be placed in a plugin in the readme from the “plugin” folder on the GitHub of our project. You will also find more information in the others readme inside the different folder from the “plugin” folder (especially for the rendering and routing files).

2. How to use a plugin

First of all, you will need to import your personal plugin in the app, and for that to import the .osf file in the app.
You can put it in a cloud space, like google drive and then download it on your phone from this cloud space. Once the .osf file is downloaded on your phone, you can open the app.

On the app, you will find the “parameters” section, where you will find “restore from a file” (at the end of the “parameters” section). Click on it, search for the plugin (your .osf file), and validate. The app will download the file, and then close. Re-open it again, and you have the plugin installed!

You can search for information about it in the “plugin” section (“extensions” in French) : you will find there your personal plugin with descriptions of it, for example how to use it in the app. You can also find the profile brought with the plugin in the “parameters” section, under “application profile”. From there, you will be able to modify the current routing and the routing parameters to use the personal routing brought with the plugin, in the “guidance parameters” section.
For example, with our shade plugin :
“guidance parameters” => “navigation type” => choose the new routing to use : pedestrian-shade_routing.
“guidance parameters” => “route parameters” => change the “hour” parameters to the current time.

Finally, we created a visual presentation of how to use the plugin with IOS. You can find it in the document's appendix, but notice it uses the screenshot from our French app. Notice also that the process is similar on Android.

3. Note about examples in our GitHub

You will find in our project's GitHub different examples of plugins in addition to the latest functional version. These examples show how to use the rendering, but it was a part we didn't finish due to technical issues.

4. Possible bug

If you do not write things in the correct syntax, and for obscure other reasons, you may encounter a bug : the application closes itself. Each time you open it, it will close instantly.
Because it is due to the personal plugin you just downloaded into the app, there is a simple way to deal with it, instead of uninstalling and reinstalling the app !

On your phone, you search for the applications storage in your parameter app. There, you should be able to find OsmAnd and its data. You can then remove the files added with your plugin (generally, only 3 files are added : the one for the rendering, the one for the routing, and the .obf map). After that, you should be able to reopen the application.

Note that we only encountered this bug when making tests with the rendering.render.xml file. If you write something wrong in the items.json file, it may break your app and this solution may not work (but we did not try it so we do not really know, it is just a warning).
About our tests that gave us this bug : the syntax is really important in the rendering.render.xml. Writing something incorrectly may not do anything (the app just won't take your rendering file into consideration, and your map will not change) or it may break the app. Also we found that modifying the renderingAttribute "gpx" is not a good idea, as it also breaks the app (test with OsmAnd version of January 19, 2026, on Iphone).

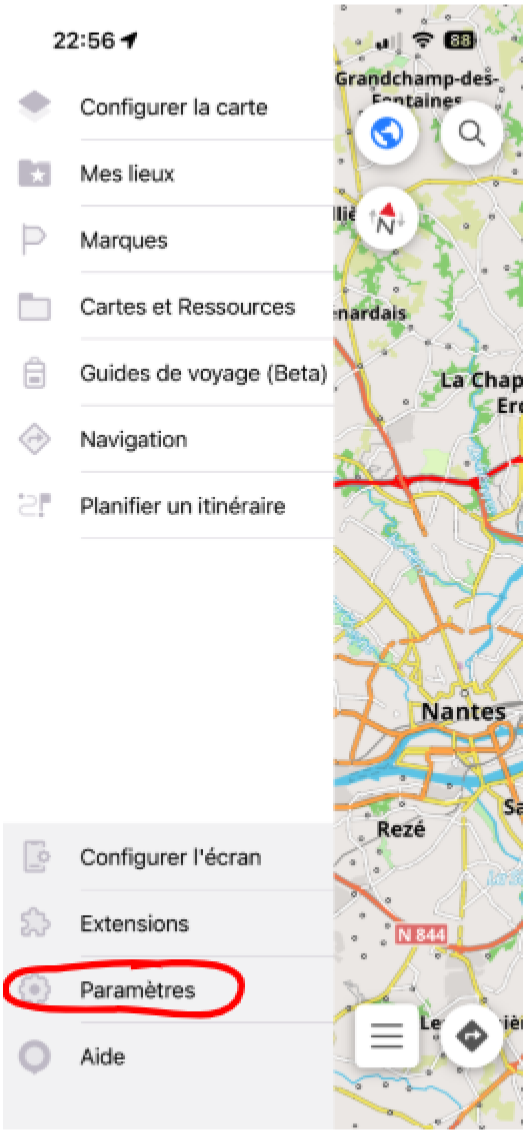
5. Appendix

Step by step visual guide of the use of a plugin with IOS

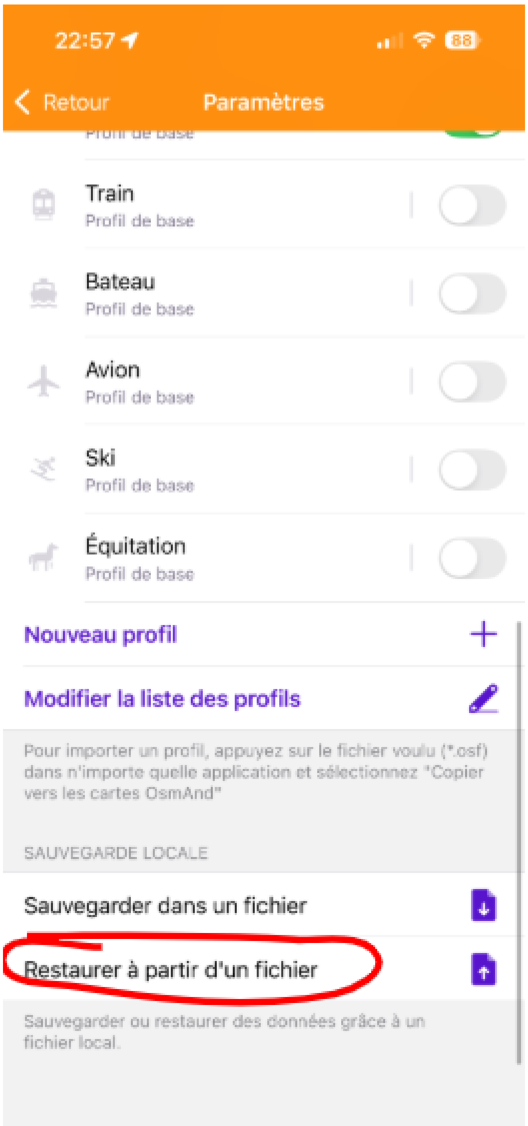
Note two things : it uses the screenshot from our French app, and the process is similar on Android.

Once you have downloaded the .osf file on your phone, open the app. It should look like this screenshot.

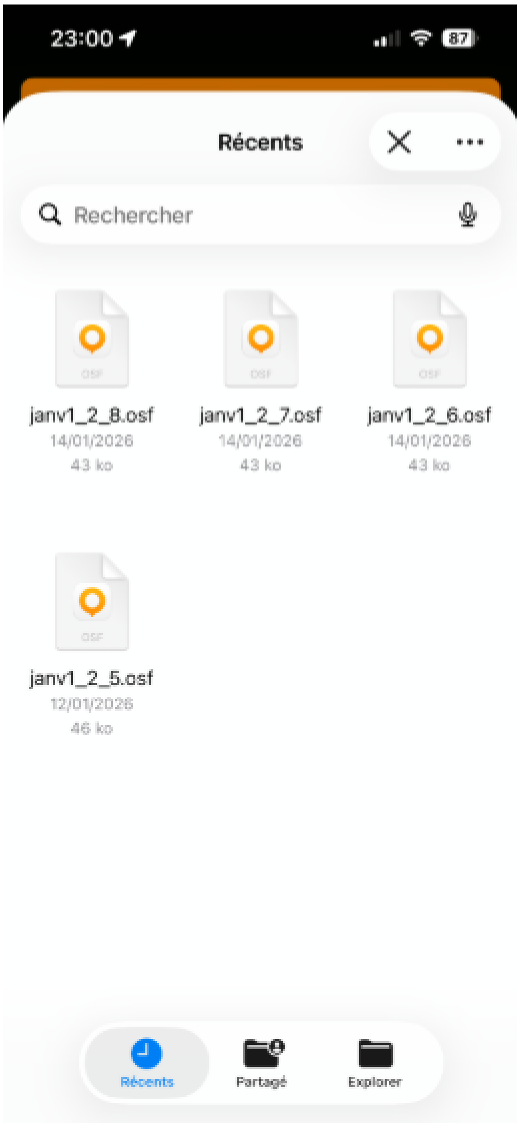
Then, you can click on the menu bar, circled in red.



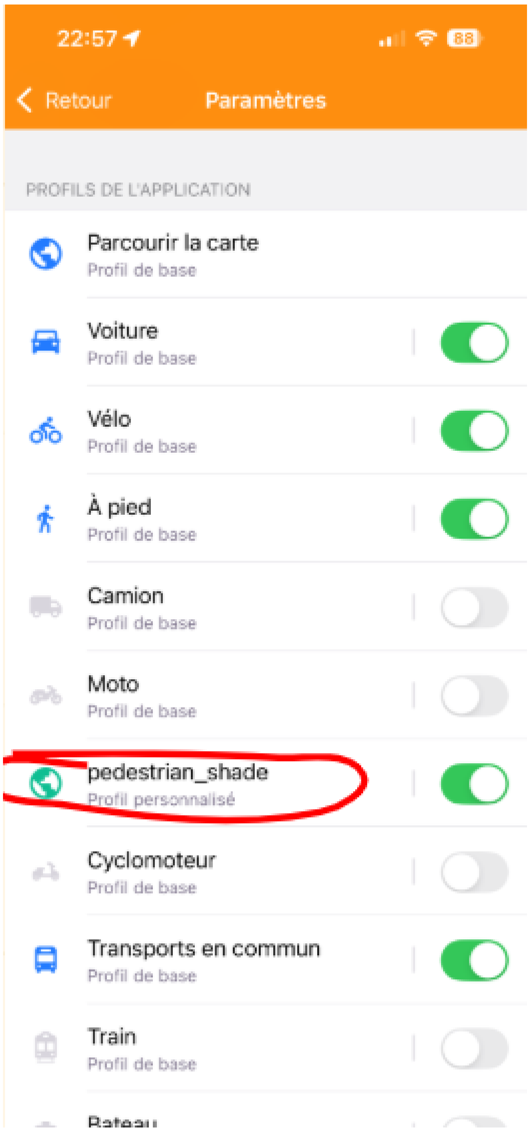
You will find the “parameters” section. Click on it.



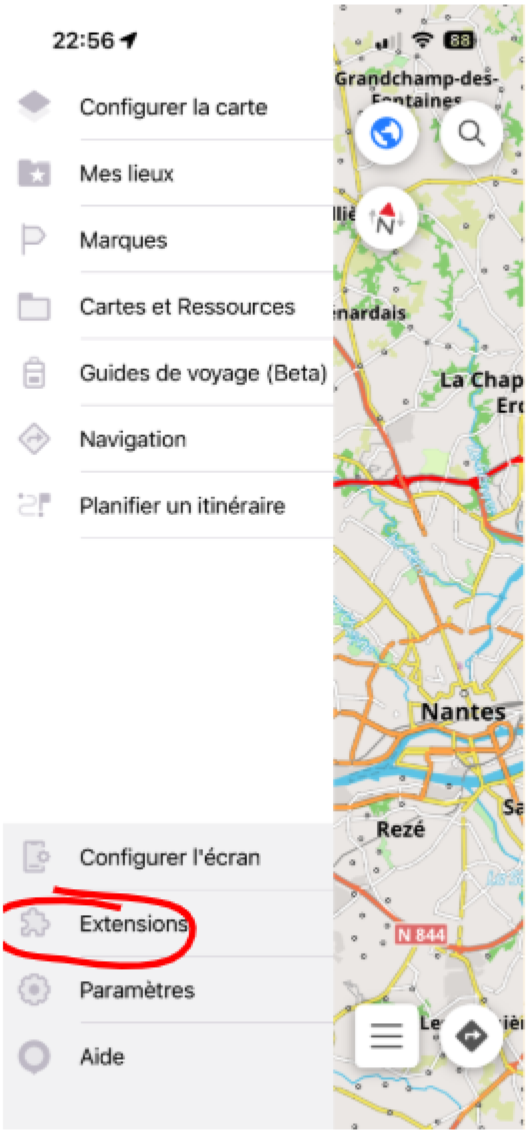
At the bottom of the section, you will find “restore from a file”. Click on it.

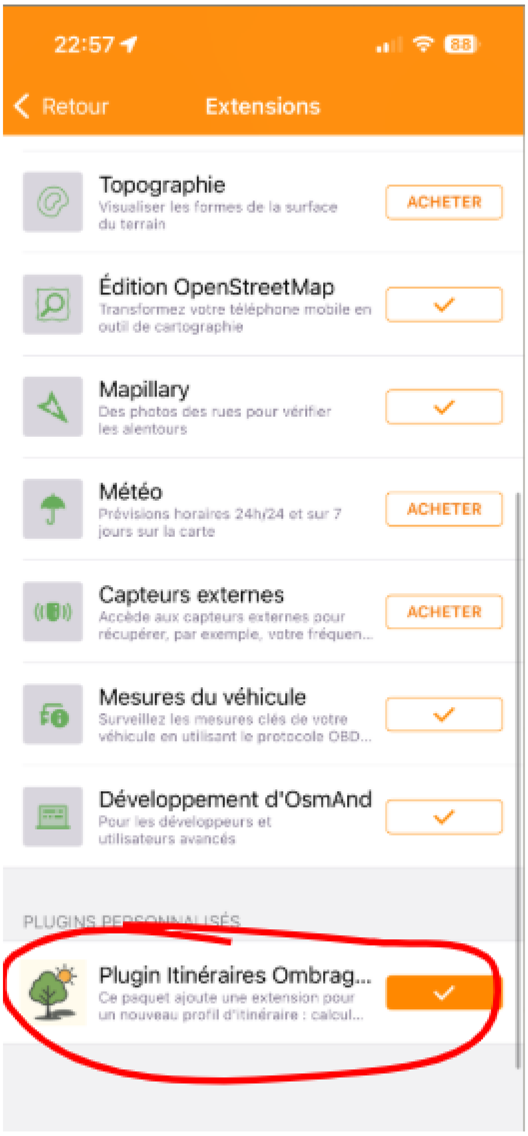


Here, you will be able to select the .osf you want to import in the app.
Select it and validate.
The app will download the file, and then close. Re-open it again, and you have the plugin installed!
After that, you can come back to the "parameters" section and see your new personal profile, under "application profile":



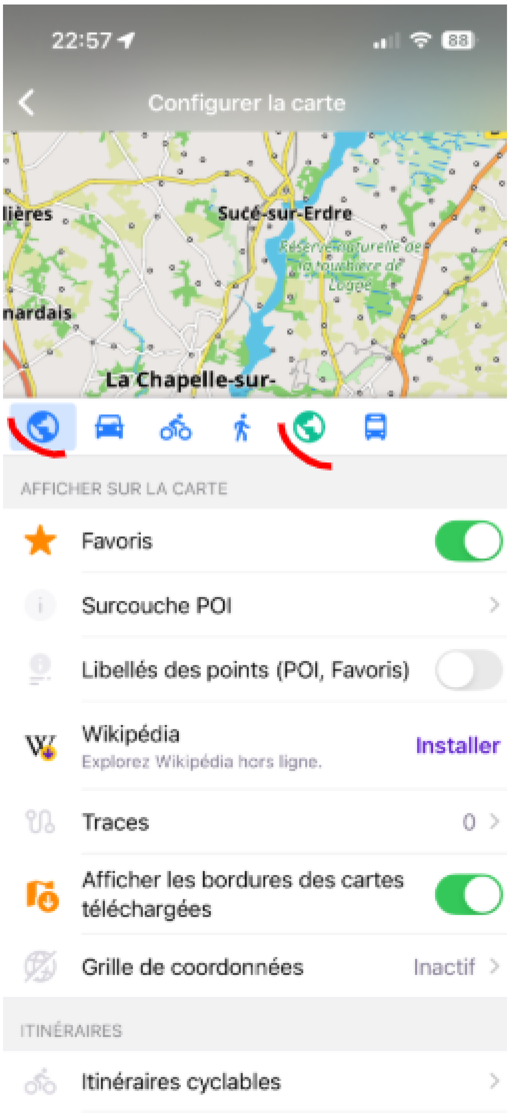
You can also find general information about your plugin, such as its descriptions, in the “plugin” section from the menu bar.


22:56
88
Configurer la carte
Mes lieux
Marques
Cartes et Ressources
Guides de voyage (Beta)
Navigation
Planifier un itinéraire
Configurer l'écran
Extensions
Paramètres
Aide
Grandchamp-des-
Nantes
Rezé
N 844

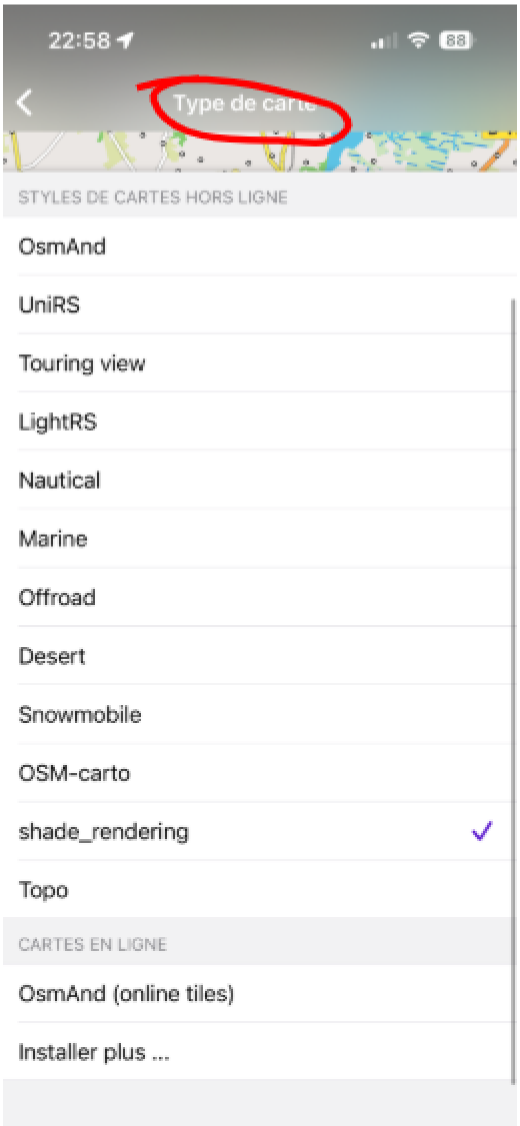

22:57
Retour
Extensions
Topographie
Visualiser les formes de la surface du terrain
ACHETER
Édition OpenStreetMap
Transformez votre téléphone mobile en outil de cartographie
Mapillary
Des photos des rues pour vérifier les alentours
Météo
Prévisions horaires 24h/24 et sur 7 jours sur la carte
ACHETER
Capteurs externes
Accède aux capteurs externes pour récupérer, par exemple, votre fréquen...
ACHETER
Mesures du véhicule
Surveillez les mesures clés de votre véhicule en utilisant le protocole OBD...
Développement d'OsmAnd
Pour les développeurs et utilisateurs avancés
PLUGINS PERSONNALISÉS
Plugin Itinéraires Ombrag...
Ce paquet ajoute une extension pour un nouveau profil d'itinéraire : calcul...

If there is personal routing or personal rendering in your plugin, here is how to activate them.
For the rendering :
Go to the map section:

Here, you can find your personnel profile. Just select the profile you want to use.

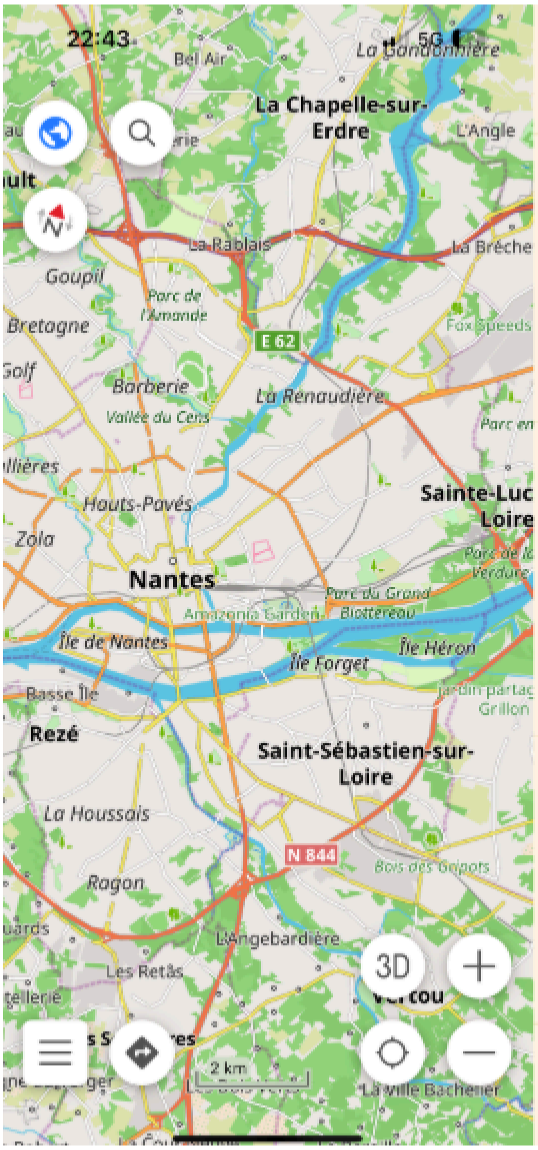


Then search for the “itinerary” or “route” section, and select “map type”.

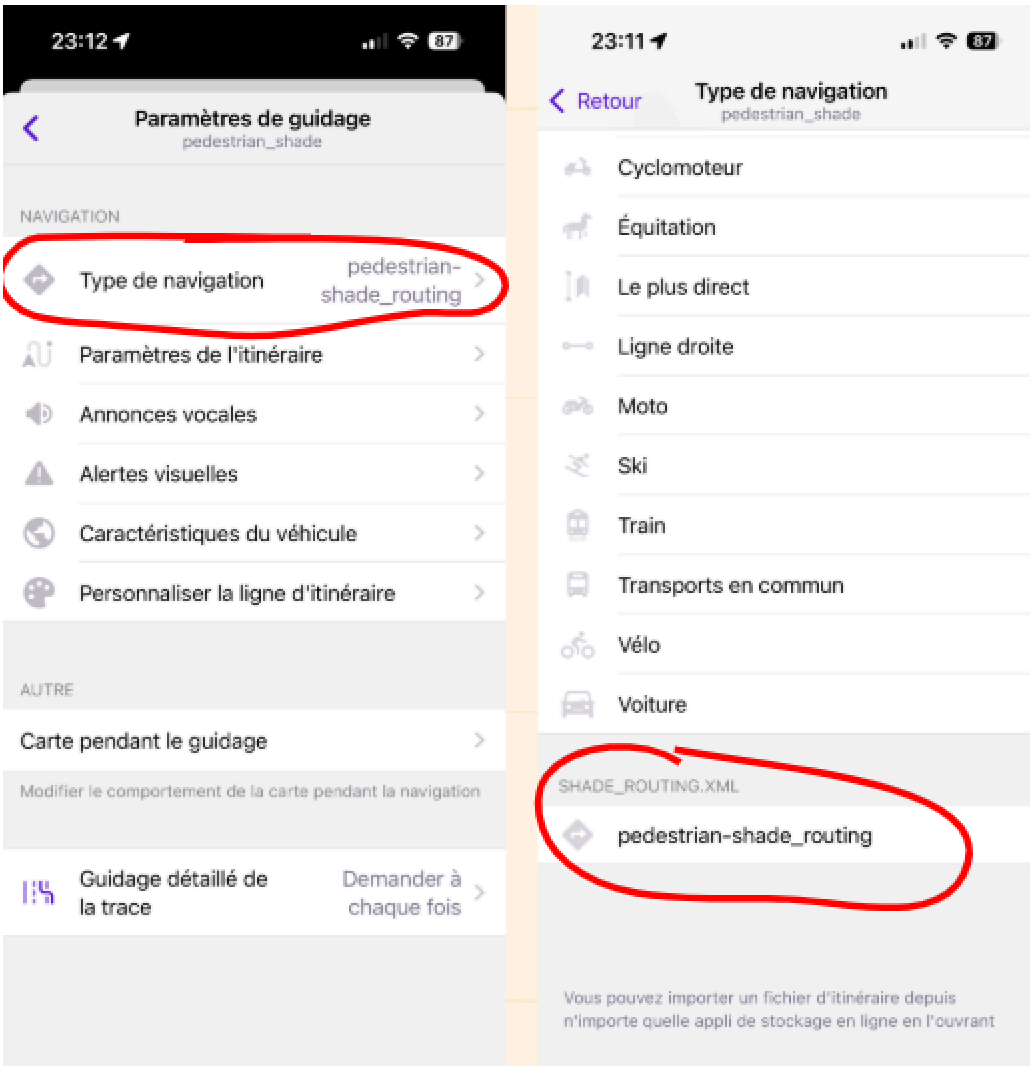


Here, you would be able to select the type of map you want to use, for example the map with shade information which uses the shade_rendering.xml file.

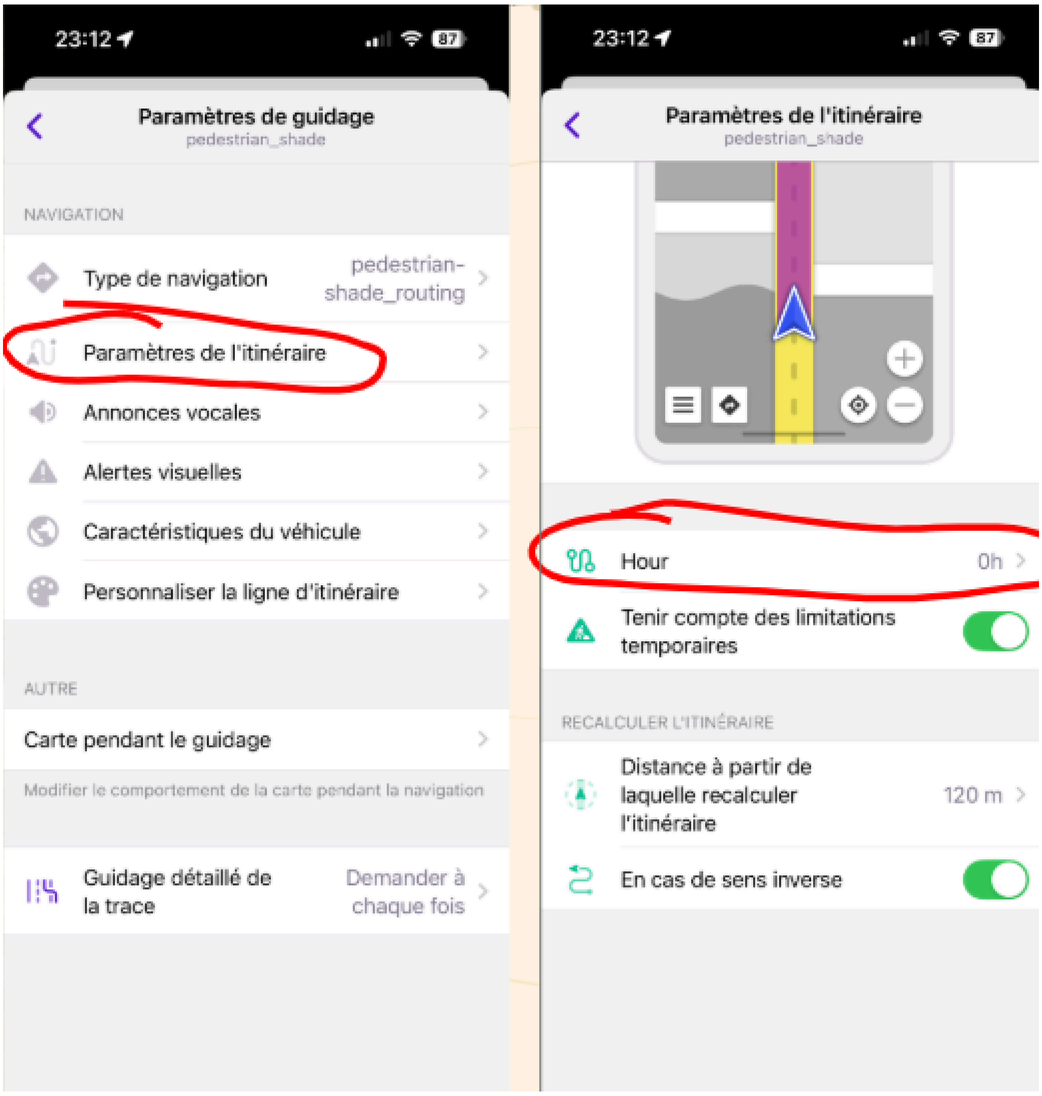
You should then see a difference between the default map and the one you selected. Here is a small example where we selected a type of map that colour several roads in red.



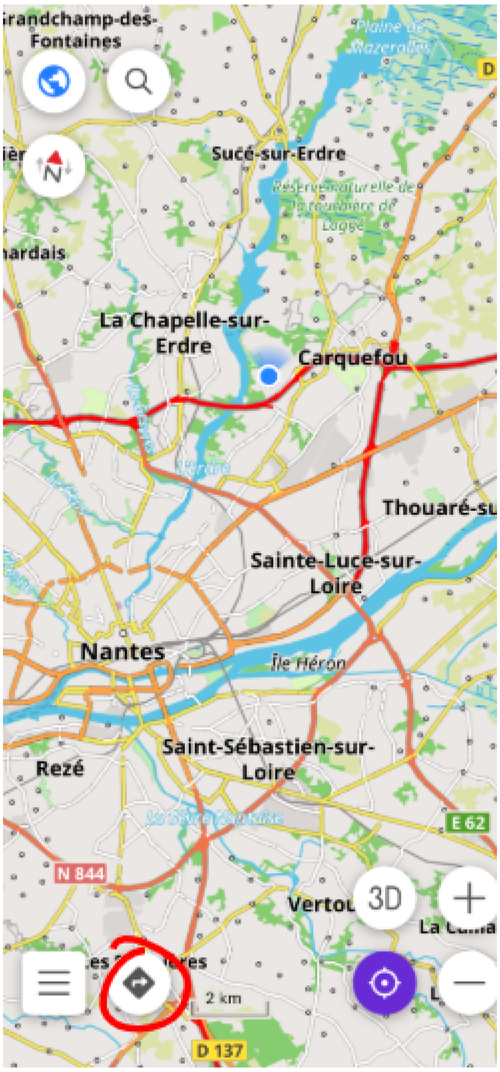
For the routing :
Go back to the “parameters” section, and select your personal profile.
We will use the example of the pedestrian_shade profile (cf. above).
Once it is done, go to the “guidance parameters” section.
“guidance parameters” => “navigation type” => choose the new routing to use : pedestrian-shade_routing.



“guidance parameters” => “route parameters” => change the “hour” parameters to the current time.



Once it is done, you can click on the “route” button, select your profile, and try to plan a route.



You can also try to plan the same route (from point A to point B) and change the profile to see the difference in route depending on the routing used.